

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) With a DB plan, the company carries the risk, whereas with a DC plan, the members carry the risk. [0.5]
- Therefore, if experience under a DB plan is good the company benefits from reduced contributions. If experience is good under a DC plan, the members will receive increased benefits. [0.5]

Differences between funding a DB and a DC plan

The DB contribution rate is a global rate, based on a weighted average of all the individual members' contribution rates. [0.5]

A DC scheme has no cross-subsidies as it is individually funded. [0.5]

Where interest rates exceed salary increases, the cost of the pension provision increases as the age of the member increases. [0.5]

Hence the DB contribution rate will provide less than the cost of accrued benefits for older than average members, but younger members will be better off. [0.5]

The contribution rate calculated will be based not only on the normal retirement age but also on the probability that members will die, withdraw or retire early. [0.5]

The rate will depend upon the funding method and assumptions used to calculate the contribution rate. [0.5]

The assumptions used to calculate the contribution rate may be deliberately optimistic or prudent depending on the aims and objectives of the various parties. [0.5]

The actual contribution rate paid will depend on the amortization of any surplus or deficit already in the scheme. [0.5]

Implications for benefits of the two different methods

Under a DC scheme the size of the pension benefits depends directly upon the experience of the scheme. Experience may be very different from that assumed in their funding bases. [1]

For example, members may have suffered poorer investment returns, had higher annuity rates, has higher salary increases or incurred greater expenses compared with that assumed. [0.5]

You would expect members retiring now may have been older than average at the time of the switch. Therefore, you would have expected the contribution rate to be insufficient to provide the benefits they would have received in the DB scheme. [1]

Even if these members were of an age similar to the weighted average of the scheme at the time of the switch, the contribution rate may still have been inadequate because: [0.5]

- It was not a prospective benefits method and therefore did not allow for the future service and ageing of these members, e.g. Projected Unit Method [0.5]

- It did not allow for future projected earnings, e.g. the Current Unit Method, which only allows for current salary or current salary plus some statutory increases [0.5]
- It allowed for an element of surplus, which would have reduced the future service contribution rate below that required to finance future benefits. [0.5]

The DB contribution rate would have allowed for the surplus that may have arisen from withdrawal or other contingencies. For a DC scheme there are no such surpluses. Hence the benefits under the DC scheme may be lower than what would have been provided under the DB scheme. [1]

Benefits taken may have been of a different format than that under the original scheme. For example, it may allow for greater increases in retirement. [0.5]

The value of the benefits may be similar, but the initial level of pension will be less. [0.5]

[Max 10]

ii) Disadvantages to employer

Risks

The following post-retirement risks would fall on the employer:

- investment risk, i.e. the risk that investment returns are lower than expected [0.5]
- longevity risk, i.e. the risk that members live longer than expected [0.5]
- expense risk, i.e. the risk that expenses are higher than expected [0.5]
- uncertainty regarding contingent dependants' pension payments, e.g. whether they are paid and how long dependants will live. [0.5]

Annuity rate

There may be high administration costs, in setting up and reviewing annuity rates. [0.5]

There is a risk that members will feel they are being treated unfairly if annuities are poor value compared with those available from the market. This could result in ill-feeling and bad press. [0.5]

There is a risk that some members will feel they are being treated unfairly if plan annuities are more generous for some ages than others, or guaranteed at some ages and not others or different for the different genders. Also, when rates change, members currently retiring may feel that they are being treated unfairly compared with members who retired in the past. [1]

Investment

Investment options may be constrained, e.g. investments may need to be held to match the annuity paid. [0.5]

Practicalities

The risk of the uncertainty of changes in taxation and legislation falls on the employer. [0.5]

Disadvantages to the member

Size of pension

There is a risk that members may not receive as favourable annuity rates from the employer as from the market...

[0.5]

... for example, the employer may not review the annuity rates frequently, or may smooth between ages and/or genders.

[0.5]

Security

There is a risk that there is adverse experience in the plan or that the employer offers generous terms and a deficit arises, resulting in pensions not being paid in full which may be due to a possible wind-up of the pension plan.

[1]

Flexibility

If annuities are provided from the plan then they may be of a set form, whereas purchase from an insurance company may give the member greater choice

[1]

[Max 8]**[18 Marks]****Solution 2:**

i)

Lack of private provision

The lack of private provision may be due to:

- concerns about the security and risks of such provision [0.5]
- a lack of knowledge about the importance of making provision [0.5]
- a lack of provision leading to a lack of competition in the market and high costs, further reducing private provision [0.5]
- apathy and the belief that the State will provide sufficient benefits [0.5]
- low wages, so no spare money to make savings [0.5]
- a lack of tax and other incentives (relative to other savings vehicles) [0.5]
- a lack of incentives for employers or other providers to offer schemes [0.5]
- lack of paternalism by employers, e.g. if the workforce is highly mobile [0.5]
- over-regulation of employer provided schemes, making employers less inclined to offer them, due to the time and costs associated with compliance [0.5]
- over-regulation of private providers and insurance companies, e.g. on selling, solvency margins etc. [0.5]
- High charges (for personal pensions or occupational schemes). [0.5]

[Max 5]

ii)

Tax concessions

The tax concessions could apply to:

- employer contributions – deducted from profits before deduction of any corporate tax [0.5]
- employer contributions – individual not taxed on them, i.e. not classed as taxable benefit or perquisites [0.5]
- employee contributions – deductible for tax purposes [0.5]
- alternatively, all contributions could be subject to a lower level of tax than otherwise (e.g. than earned income or profits) [0.5]
- investment income to receive beneficial tax treatment [0.5]
- investment growth (realised and unrealised gains) to receive beneficial tax treatment [0.5]
- pension to receive beneficial tax treatment, e.g. not taxed as income [0.5]

	• cash lump sum to receive beneficial tax treatment, e.g. not taxed as income [0.5] • death benefits to receive beneficial tax treatment, e.g. not taxed as income [0.5] • Benefits (pension and/or lump sum) subject to a lower level of tax than otherwise (e.g. earned income). [0.5]	[Max 4]
iii)	Defined contribution retirement provision could be via an employer provided scheme or private provision. The State may choose to impose different regulations in some areas for the two approaches. [0.5] Areas where regulations may be imposed: Operational – there may be restrictions on how such arrangements are set up, i.e. under trust or contract. There may be more protections if the scheme is set up under trust, e.g. assets held separately from the sponsor. [1] Maximum contribution level (sponsor and/or employee) – there may be restrictions on the level of contributions, in particular if tax incentives are given. The maximum level of contributions may be age-related to reflect the cost of accrual increasing with age (provided investment return exceeds salary increases). [1] Minimum contribution level (sponsor and/or employee) – a minimum level of contributions could be prescribed to try to ensure adequate benefit provision. [0.5] Taxation of contributions, investments, benefits – regulations may be imposed relating to the tax treatment of contributions, investment income, pensions and any lump sum cash payment. [1] Minimum level of benefits – may be required to target a minimum level of benefits, or schemes could be required to provide some form of defined benefit underpin. [0.5] Type of investments – restrictions on investment choice could be imposed, e.g. limits on self-investment or the use of equities or derivatives. [1] Form of benefit provision – the form of benefit provision could be prescribed, e.g. require the pension to be index-linked. Regulations may allow some amount of the benefit to be commuted for cash. [1] Age at which benefits may be taken – minimum and maximum ages of retirement may be prescribed. [0.5] Provision of partner benefits – may require provision of a partner’s pension if the member has a partner. [0.5] Marketing of private benefit provision – regulation could be imposed on the quality of promotional material and the form of any illustrations. [0.5] Level of charges – legislation could impose a maximum level of charges by the pension provider, e.g. insurance company or pension fund manager [0.5] Disclosure of information – legislation may outline the frequency and nature of disclosure of information. [0.5]	

Advisors – requirement to take advice or for the advisor to be properly qualified and/or regulated.

[0.5]

Regulation of finances of providers – e.g. solvency margins for insurance companies.

[0.5]

[Max 7]

iv)

Advantages of the proposal

- if combined with reducing State benefits it can control State costs
- general inertia means many people may remain in the scheme even though an opt-out option is offered, thus increasing provision and overall savings
- the opt-out option is more politically acceptable than if the arrangement is compulsory
- easily copes with contribution holidays, career breaks, change of employment
- requiring employer contributions should help ensure a non-trivial amount of money is saved, leading to meaningful benefits
- requiring only modest employer contributions should not be too onerous a burden on employers
- State regulated and therefore expected to be secure
- a funded arrangement so may have positive impact on investment markets, creates demand for investments
- offering investment choice enables individuals to tailor investments to their risk appetite and circumstances
- DC in nature and therefore transparent and easy to understand
- this may also allow flexibility over the form and timing of the benefit
- funded and therefore likely to be more secure
- there should be benefits from economies of scale, i.e. lower administration and commission costs

Disadvantages of the proposal

- does not cover long-term unemployed, long-term carers, sickness
- not compulsory, and so does not ensure employed individuals have private provision
- unpopular as it may “appear” to be compulsory
- may have a negative impact on existing private provision, e.g. will existing final salary schemes be more likely to close?
- does not guarantee the pension since DC scheme and the investment risk and longevity risk falls on the individual
- e.g. there is a risk of poor investment returns over the period to retirement or a fall in value at retirement
- e.g. there is a risk of the conversion terms at retirement being worse than expected due to longevity improvements
- individuals may feel they do not have sufficient knowledge to make investment choice, or may just make inappropriate choices ...
- ... leading to the need for a default option or individuals paying for investment advice
- contributions made may not be sufficient to provide an adequate pension
- may be unpopular with employers as required to contribute
- there may be a need to offer some guarantee or underpin to the pension, with associated costs
- if the State offers tax incentives as encouragement then there is an associated cost
- a significant change to the system with associated costs, e.g. education will be

- required and a new administration system
- additional regulation will be needed
 - there may not be a large enough market to support the buyout of benefits at retirement
 - may not lead to a higher level of savings, just a change of savings vehicle.
- [0.5 per point] **[Max 10]**
- v) Pros:**
- popular with employees (if it is offered as an option, this has a high take-up rate, even if the terms are worse than cost neutral) [0.5]
 - better value than pension for members who die soon after retirement (e.g. for those in ill health at Normal Retirement Age) [0.5]
 - flexible, e.g.: [0.5]
 - employee can spend it or save it or take half of it [0.5]
 - can be used to pay off debts (e.g. a loan) which may be attracting high interest charges [0.5]
 - may have tax advantages relative to taking the benefit as a stream of income [0.5]
 - if employee wants only a pension they can buy an annuity or a property yielding rental income. [0.5]
- Cons:**
- if offered as an option in exchange for pension then possible selection against scheme, i.e. those in poor health commute, healthy don't [0.5]
 - lump sum often squandered [0.5]
 - perhaps badly invested even if not squandered [0.5]
 - may make pension appear mean, i.e. poor publicity for company [0.5]
 - may require pension scheme to realise assets, perhaps at a poor time (only likely to be a problem for small or mature schemes) [0.5]
 - Members who live longer than average lose out. [0.5]
- [6]**

Solution 3:**[32 Marks]**

- i) Method of provision**
- If a benefit scheme is being discontinued, a number of options may exist for the provision of the outstanding benefit payments. Six examples are covered below:
- gradual removal of liabilities through temporary continuation
 - transfer the liabilities to another pension scheme with the same sponsor
 - transfer the funds directly to the beneficiary to extinguish the liability
 - transfer the funds to an insurer to invest and provide a benefit
 - transfer the liabilities to an insurer to guarantee the benefits
 - Transfer the liabilities to a central discontinuance fund.
- [0.5 marks for each point] **[3]**
- ii) Transferring funds to beneficiaries**
- The issues for the trustees:
- Security of the scheme – the ability of the scheme to provide benefits and in the given situation it appears safe to provide the benefits in hand. The impact on the security of the scheme for the remaining members of this which in this case appears very few given the Government intention to liquidate the company itself, or will members receive reduced transfer values? Are more funds available if there is a deficit? [0.5]

	• Use of the funds – how will members use the funds? Will the funds be squandered, or will there be any restrictions on the use of the funds? [0.5] • Fairness – will this option result in all categories of members being treated equitably. Are there any winners or losers against other possible options? The other possible options may be ruled out as the company itself is in liquidation stage and reaching out to the insurer may take time [0.5] • Scheme design – what allowance is made for complex and discretionary benefits under this option and the other options? Are material individual characteristics taken into account in some options? What is their worth? [0.5] • Investment policy – assets will need to be realised and this could result in uncertainty where there are volatile assets and potentially high dealing costs. Is it a good time to realise assets in the market as the economy overall is not good? [0.5] • Risk – DB risks will be removed from the scheme and the risks will be transferred to the beneficiaries for those who take up this option. [0.5] • Advice – is there a need to provide advice to members as to whether to take this option and, if they do, subsequently what to do with the funds? [0.5] • Legislation, administration and communication considerations. [0.5]
	[4]
iii)	Setting the terms
	<i>Regulation</i>
	Regulation may dictate the approaches permitted on discontinuance. [0.5]
	If this approach is permitted then there may be legislation that will affect the terms. For example, anti-discrimination legislation may mean that unisex factors must be used or the basis may be prescribed to some degree. In addition, there may be minimum and maximum amounts that can be paid. [1]
	<i>Security</i>
	The terms should be set so that the exercise of this option should not jeopardise the financial security of the scheme.
	Terms could be set using the ongoing funding basis so that the exercise of this option would have no effect on the ongoing funding level. Alternatively, a best estimate basis could be used so as to be cost neutral in terms of the actual cost of benefits. [1]
	<i>Funding position</i>
	The funding position of the scheme should be considered, if it is underfunded and the employer is not in a position to make up the shortfall then the funds to be transferred to the beneficiaries will be reduced. [1]
	If the scheme is in surplus, then ownership of the surplus will need to be considered and, in particular, whether the funds to be transferred will be augmented. [0.5]
	<i>Selection</i>
	The terms for this option should be set to minimise any risk of selection, eg those in poor health could be more likely to take the fund if the terms do not reflect their health status however this selection effect does not appear to be relevant here as the situation of the company is really critical. [0.5]

	Sponsor's view The sponsor's view needs to be considered. For example, the sponsor may wish to encourage members to take this option so that the DB risk is removed and therefore the terms could be set to be generous.	[1]
	Fairness The terms should be set to provide value for money in respect of the DB benefits being given up. The terms should also be fair to members who take, and who do not take, this option. This suggests cost-neutral terms. Equity should be considered against the terms offered on any alternative options.	[1] [0.5]
	Market conditions The terms are likely to reflect current market conditions as liabilities are being extinguished and this is a short-term valuation. Thus the current investment policy will need to be considered.	[0.5]
	Simplicity The terms should be simple so that the option is simple to administer and communicate. This may lead to terms which do not reflect individual circumstances; this could increase the risk of selection.	
	Expenses The trustees need to agree who will meet the expenses of offering this option, and therefore whether the terms need to include an expense allowance.	[0.5]
	Take up rate If it likely that most members will take up this option, then selection is less of a risk which actually is the case here. In addition, if assets need to be sold, the terms may need to reflect the cost of a quick bulk sale of assets.	[0.5]
	If few members are likely to take up this option then individual terms (to reflect the selection risk) or broad brush terms (to minimise the marginal expenses of the exercise) may be used but this may be ruled out at this stage.	[0.5]
	Use of funds The terms may reflect how the funds would be used. For example, if members are required to purchase a deferred or immediate annuity with an insurance company then the terms may reflect the terms upon which such contracts are available.	[1]
	Options The trustees need to consider how to allow for options that are not cost-neutral, eg commutation factors. Should it be assumed that all/none take-up the options or allow for a proportion?	[0.5] [Max 8]
iv)	The question still remains if it's sold and the new company has a defined benefit scheme and till then the future of the employees is uncertain.	[0.5]
	Possible advantages would include:	

- cashflow synergy between the two schemes, ie for an immature scheme, contributions in are greater than liability cashflows out. Consequently these “extra” contributions can be used to cover the liability cashflows for the discontinued scheme. This avoids simultaneous buy/sell decisions, resulting in reduction in dealing costs. P [1]
 - fewer investment constraints caused by liability matching requirements [0.5]
 - larger fund gives opportunities to invest in areas/assets not otherwise available [0.5]
 - sharing of any fixed costs resulting in lower overall costs. [0.5]
- [3]**
- v)** The major disadvantages are:
- to the extent that discontinuance arises because of company liquidation, the levy is likely to be required at times when companies can least afford to pay, eg in times of recession. This depends upon the strength of the sponsor covenant
 - Well-funded schemes may end up subsidising poorly funded schemes
 - it may encourage scheme managers to pursue less prudent funding and/or investment strategies
 - if the fund only guarantees to cover a certain percentage of the liabilities, then the level of security that is achieved is lower than members’ expectations. [3]
- vi)** How accounts can be used to check valuation data
- The accounts provide an independent check to validate the valuation data. [0.5]
 - The sponsor’s and members’ contributions should be consistent with both salary roll and the contribution rate. [1]
 - Sponsor’s and members’ contributions should be consistent, eg if members pay 5% and sponsor pays 20% then the sponsor contributions in the accounts should be four times the member contributions. [0.5]
 - Total pensions paid (with adjustments for deaths and new pensioners) should be consistent with the pension payroll in the valuation. [1]
 - Membership numbers should tally.
 - Membership movements should tally with new pensions, new cash sums, death benefits paid and leaver benefits paid. [0.5]
 - Asset data should be consistent with investment income. [0.5]
 - The value of assets should correspond with the investment managers’ performance, contributions paid, monies paid out and the latest valuation of assets. [0.5]
- [5]**
- vii)** Disadvantages of summarised data
- The absence of such detailed data may mean the scheme cannot:
- produce a valuation result which is as accurate as possible [0.5]
 - allow the benefits to be valued accurately, eg a money purchase underpin [0.5]
 - enable individual membership data to be checked, eg reasonableness checks, cross-checks against previous years’ data, reconciliations [0.5]
 - investigate the reasons for errors [0.5]
 - Analyse past experience in detail, perhaps leading to the choice of the most suitable assumptions. [0.5]
- [Max 2]**

- viii) When it may be necessary to accept summarised data The circumstances in which it might be necessary to accept such data are those in which the advantages of summarised data (ie simplicity, speed and cost of calculation) outweigh the disadvantages above (eg accuracy). [1]
- This may be true of any valuations that are carried out between the formal funding valuations. (The purpose of such a valuation may simply be to check that there have been no major deviations from expectations.) [0.5]
- It may also be true in the case of advice relating to a company purchase where detailed information may not be available early on in the negotiations. [0.5]
- A further example may be calculations for the company accounts for years between formal funding valuations. [0.5]
- [Max 2]**
[30 Marks]

Solution 4:

- i) Reasons for analyzing surplus
- Analysing surplus is part of the “Monitoring of experience” stage of the actuarial control cycle. It is one of the regular reviews that lead to an amendment of objectives and assumptions in the light of experience. [0.5]
 - To provide an independent check on the valuation result. [0.5]
 - To indicate those features of the past experience most likely to recur. [0.5]
 - To measure the relative financial importance of departures from what was expected. [0.5]
 - To provide useful background information to help when communicating to the client the reasons for changes in the funding level and contribution rate. [0.5]
- [Max 2]**
- ii) Mortality before retirement
- Higher than expected pre-retirement mortality increases the death benefits payable. [1]
 - However, the effect on the surplus depends on whether the benefits are insured. [1]
 - If the death benefits are fully insured, the surplus would be increased by the release of the reserve held in the gratuity scheme for the member’s retirement benefits. [1]
 - If the death benefits are only partially insured, ie insurance only covers the death benefits in excess of the reserve held for the member’s gratuity benefits, then the mortality experience will be cost-neutral. [0.5]
 - If the death benefits are not insured, then the effect on surplus will depend on the relative values of the death benefits payable and the reserves held in the scheme for the member’s total benefits. [1]
 - The largest surplus will arise from deaths of members with long service, for whom the reserves in the scheme are large compared with the death benefits. [0.5]
 - The largest deficits will arise from deaths of members with short service (eg young members), for whom the reserves in the scheme are small relative to the death benefits. [0.5]
- [Max 2]**

iii)	- Assets are valued using a current bond yield. This may have no relation to how the assets (if any) are invested and could vary between the two valuation dates. - Contributions (if any) to the arrangement could be made on a basis which bears no direct relation to the cost of accruing benefits - For example there could be a fixed percentage of payroll or determined using a basis fixed by statute - The cost of benefits could be different from those assumed For example if they are related to salary — salary growth at the company could be different from that assumed in the standard (prescribed by legislation) table or an individual may receive, for example, a promotion which increases the cost of their benefits or there may be options, either for the member or the employer which result in a higher rate of benefit being paid (for example, ability to retire early). - Either because of changes in market conditions or because the assumptions prescribed bear no relation to prevailing costs. - The benefit formulae could have been changed - The actual investments, (if any) of the scheme could have performed quite differently from that assumed. - For example if they are invested in property or different bonds from those specified in the legislation, actual returns would be different. - Costs of administration etc., if these are met from the arrangement could be different from those assumed or indeed they may not be assumed. - Normal statistical differences could occur, for example, mortality rates; actually experienced etc could be different from those assumed. - There may have been a significant change in the membership of the arrangement, for example a disposal or an influx of new employees without a consistent alteration in the rate of contribution or assets. - There could be an error in the calculation or data provided.	[1] [1] [0.5] [2] [0.5] [0.5] [0.5] [0.5] [0.5] [0.5] [1] [0.5] [9]
iv)	Asset liability Modeling : Information Required: - Precise objectives - ensure appropriate data/method used - define risk that is being measured - define time period to be considered - example of risk (e.g.) 95% probability of funding position over 100% for 10 years) - example of objective(e.g.) highest possible return meeting risk criteria) - time horizon important - credibility of results are time dependent sponsor may be considering short timescale(e.g. 5 years) - so model needs short term assumptions - often easier to set assumptions for longer period - need all of data for regular funding review- in addition need terms or any options/guarantees - need funding method - and assumptions used to set funding target - need number of simulations - e.g. need 10,000 to meet 5th percentile criteria	[0.5] [0.5] [0.5] [0.5] [0.5] [0.5] [0.5] [1] [0.5] [5]

[20 Marks]
